Bitter Pills: Sourcing under Compressed Timelines and the Cost of Compliance in Brazilian Public Health Procurement*

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Abstract

Judicial enforcement is essential for the rule of law, yet the *design* of enforcement can generate economic inefficiency. We provide bid-level evidence on the procurement cost of court-mandated pharmaceutical purchases in São Paulo, Brazil (2009–2019). With item, time, and buyer fixed effects, urgent purchases carry negotiated prices 5.4% higher and attract -5.4% fewer bidders than ordinary purchases of the same item; they also succeed 2.1 pp more often. Within urgent purchases, a parallel administrative-request channel that shares all planning constraints but carries no penalty for officials offers an institutional contrast. The naive "under the gun" (UTG) gap between litigated and administrative urgent purchases of the same item is large (29.5%), but it is contaminated by selection: the SES/SP scientific committee admits cases

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on cost-effectiveness criteria. Manski-Lee bounds put the selection-corrected UTG at $[15.9\%, 21.1\%]$. The economically interesting finding is what produces the gap. Within firm-buyer-item triples observed in both regimes, the same firm sells the same item to the same buyer at essentially the same per-unit price (0.035, SE 0.041, NULL): there is *no* within-firm markup. The price margin operates through demand fragmentation (smaller orders forgoing bulk discounts) and through equilibrium changes in *which* firm wins each contract. Accountability-driven sourcing constraints, not price discrimination, are what make compliance expensive. A bounded welfare calculation places the per-unit-price cost of the litigated regime at \$27.8 M per year on the \$300 M of annual litigated spending.
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JEL: D44, D73, H51, H57, I18, K41

1. Introduction

A judge orders the government to buy a cancer drug within 1–10 days. The procurement officer has no time to aggregate demand across other purchases, no time to negotiate with multiple suppliers, and faces personal fines if delivery fails. The drug arrives. The natural question is why it costs more—and whether the cost margin is best understood as suppliers extracting a markup from a constrained buyer, or as a constrained buyer ending up with a different, more expensive supplier altogether.

The answer matters for procurement design. If urgency raises prices through within-firm markup, the policy lever is on contract terms— deterring opportunistic pricing under sanction exposure. If urgency forces the buyer onto a different point of the supplier distribution, the lever is on demand aggregation and supplier search. The two readings are observationally similar in standard data. They are observationally distinct only in data, like the bid-level data we use here, that allow comparison across regimes within the same firm-buyer-item triple.

This paper provides such evidence using all pharmaceutical procurement by the São Paulo State Department of Health (SES/SP) from 2009 through 2019: 479,330 purchase-offer-item observations conducted through the state's electronic procurement platform (BEC). Court orders and administrative requests partition urgent purchases into two types that share all planning constraints—compressed timelines, small quantities, diverted budget resources, the same BEC platform and bidding procedures—but differ in penalty exposure. Litigated purchases expose officials to fines, personal liability, and fund seizure if delivery fails. Administrative requests, instituted by SES/SP in 2009, carry no such penalties. Comparing these two types within the same item is what we call the "under the gun" (UTG) comparison.

The administrative subsample is itself selected by an SES/SP scientific committee on cost-effectiveness criteria. Pre-period probit evidence confirms the committee admits cases that are systematically less expensive (probit coefficient on pre-period log reference price $= -0.051$, $p < 0.001$) and more clinical-essential (SUS-formulary coefficient -0.265, $p < 0.001$); rejected cases land in the litigated sample. The naive UTG gap therefore confounds the sanction effect with selection on item economics. We bound the selection wedge formally: under Manski-Lee monotone restrictions on the committee rejection rule, the UTG gap lies in $[15.9\%, 21.1\%]$. This range remains economically meaningful; the rest of the paper asks where it comes from.

Three findings frame the answer.

First, the headline urgent-vs-ordinary premium is 5.4% across the full sample, with the same item under compressed timelines attracting -5.4% fewer bidding firms. Urgent tenders nonetheless succeed 2.1 pp *more* often, ruling out an alternative in which urgency simply selects easier purchases.

Second, the price margin is *not* a within-firm markup. Within firm-buyer-item triples observed under both the litigated and the administrative urgent regime (1,206 triples, 4,573 observations), the Admin coefficient is 0.035 (SE 0.041, $p \approx 0.4$). The same firm sells the same item to the same buyer at essentially the same per-unit price under either regime. Whatever pushes the across-sample price up, it is not opportunistic pricing by suppliers that already serve the buyer's procurement.

Third, the bounded UTG admits an arithmetic decomposition into a mechanical bulk-discount component (admin orders are $\sim 3.3\times$ larger than litigated orders, mechanically pulling the admin-vs-lit price gap to about -0.397 log-points), the within firm-buyer-item null, and a positive supplier composition residual ($+0.103$ log-points). Same firms price the same items the same way regardless of regime. Different firms win urgent contracts under the two regimes, and the equilibrium-weighting shift is the residual we read as the sourcing channel.

Three contributions.. *First*, we provide direct within-item evidence on the price impact of court-mandated procurement, exploiting the institutional within-urgent contrast unique to São Paulo. *Second, and conceptually most novel, we distinguish the sourcing channel of procurement-cost variation from the pricing channel.* Standard intuition under accountability mechanisms relies on within-firm pricing distortion (Prendergast, 2007); our within firm-buyer-item triple evidence cleanly rejects that mechanism in this setting. The empirical signature of accountability-induced inefficiency in our data is mismatched suppliers and fragmented demand, not extorted prices. *Third*, in contrast to Coviello et al. (2018)—where slow courts *extend* procurement timelines and raise prices—we study judicial injunctions that *compress* timelines, with the cost concentrated in items where demand aggregation matters most. The two papers identify alternative institutional channels through which judicial intervention raises procurement costs.

Welfare.. Applying the bounded UTG to the \$300 M of annual litigated spending in São Paulo, with an admissibility-share calibration of 50.0%, gives a per-unit-price welfare bound of \$27.8 M per year, with the Lee-endpoint range delivering $[\$23.9\ M, \$31.7\ M]$. We report a single bounded figure rather than the stacked ranges common in the policy literature.

Position vis-à-vis the literature.. The paper sits between the public-economics literature on procurement frictions (Bandiera et al., 2009; Best et al., 2023; Bosio et al., 2022; Decarolis et al., 2020) and the institutional literature on the judicialization of health in Latin America (Wang, 2015; Ferraz, 2009; Biehl et al., 2009). It contributes to a smaller literature on bureaucratic distortion under accountability mechanisms (Rasul and Rogger, 2018; Williams, 2021; Bandiera et al., 2021) and on competition design under contractual incompleteness (Carril et al., 2026). Our methodological discipline avoids the post-treatment-bias trap (Acharya et al. 2016): we read quantity-controlled and firm-fixed-effect specifications as channel-isolating, not as direct-effect estimators, and present the within firm-buyer-item triple as the mechanism's positive evidence.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting. Section 3 presents the data. Section 4 details the empirical strategy and the selection-bound design. Section 5 presents results, organized around the sourcing-vs-pricing pivot. Section 6 concludes.

2. Institutional Background

2.1. Health as a Constitutional Right and the Rise of Litigation

Brazil's 1988 Constitution created the Unified Health System (SUS), which maintains a formulary of approved medications subject to cost-effectiveness criteria (Castro et al., 2019; Soares, 2019). Courts, however, have adopted a strict literal interpretation of Article 196 ("health is a right of all and a duty of the state"), under which any individual can obtain virtually any medication through litigation—from common analgesics to rare-disease treatments costing over US$400,000 per year. Access is inexpensive (a prescription and a public defender suffice); approximately 85% of first-instance claims are granted (CNJ/INSPER, 2019). The result is 96,000 health court orders nationally in 2017, a 913% increase in São Paulo since 2008 (Online Appendix, Figure A.1). Litigated health spending in São Paulo alone amounts to approximately US$300 million annually—nearly 5% of the state's public health budget.

2.2. The Sanction Channel: Administrative vs. Litigated Purchases

The São Paulo State Department of Health (SES/SP) manages health policy through 99 decentralized public buyer units (PBUs). All purchases—ordinary and urgent—are conducted through the *Bolsa Eletrônica de Compras* (BEC), São Paulo's electronic procurement platform, mandatory for common goods since 2007. Two competitive procedures are used: sealed-bid tendering (*convite*) and multi-round descending auctions (*pregão*). For ordinary purchases, the internal phase typically spans 30–180 days; almost all court decisions (99.94%) are enforced as preliminary injunctions with delivery deadlines of 1–10 days, compressing the internal phase to roughly one-third of the ordinary timeline.

A critical institutional feature for our analysis is the distinction between *litigated* and *administrative* urgent purchases. Both types face identical planning constraints—compressed timelines, small quantities, diverted budget resources, and the same BEC platform and bidding procedures. They differ sharply in the consequences of procurement failure. For litigated purchases, failure to comply with a court order exposes public officials to substantial fines (sometimes disproportionate to the purchase value), administrative and criminal liability, and seizure or blocking of public funds; these penalties are public information, as the court order must be referenced in the tender notice. Since 2009, the SES/SP has run a parallel *administrative-request* channel: a mechanism allowing the government to negotiate supply directly with individuals before litigation occurs. Administrative requests undergo evaluation by a scientific committee using evidence-based medicine criteria, and—crucially—failure to complete an administrative purchase carries no penalties for public officials. Between 2009 and 2019, administrative requests generated approximately 9,700 purchases, or about 6.5% of the total from court orders.

This channel structure is the institutional vehicle for our identification. Administrative and litigated urgent purchases share all execution constraints but differ only in the penalty regime; a within-item comparison between them isolates the cost margin associated with the sanction regime, conditional on a selection wedge we bound formally in Section 4. Figure 1 summarizes the three purchase types. The administrative subsample is itself a non-random subset of urgent demand—admitted by the SES/SP scientific committee on cost-effectiveness criteria—so the naive "under the gun" contrast confounds the sanction effect with selection on item economics. We address this directly: Section 4 reports Manski-Lee bounds and a Heckman parametric correction that pin down the sanction-related component of the gap.

Table 1: Purchase Types: Institutional Characteristics

	Ordinary	Administrative	Litigated
Source of funds	Dedicated budget	Diverted budget	Diverted budget
Quantity	Large	Small	Small
Delivery time	Long (30–180 d)	Short (1–10 d)	Short (1–10 d)
Threat of punishment	None	None	**Fines, liability, fund seizure**

Notes: Administrative and litigated purchases share planning constraints (short timelines, small quantities, diverted budgets) and differ only in the penalty regime. The "under the gun" comparison exploits this variation.

3. Data and Sample

3.1. Data Sources

We combine three administrative datasets covering January 2009 through December 2019: (i) bid-level records from the BEC electronic procurement platform for all SES/SP purchases of medical supplies (BEC Group 65), where each observation is a purchase-offer-item (POI) recording reference prices, bid prices, number of bidding firms, quantities, and tender outcomes; (ii) purchase-type classification, assigning each POI as ordinary, administrative, or litigated using a regular-expression algorithm applied to tender notices, which by law must reference any underlying court order;[1] and (iii)

[1] Online Appendix §A.7 documents the classifier's design and a stratified hand-labeling protocol over 500 notices (∼167 per predicted class). The regex relies on literal court-order

health litigation records from S-CODES, the SES/SP database of health-related lawsuits, from which we derive SUS formulary status and injunction enforcement data. The binary outcome *tender success* equals one if the POI closes with an accepted winning bid within its procurement order and zero otherwise; re-tendered items enter the data as new POIs and are not counted retroactively as successes. All continuous variables are winsorized at the 1st and 99th percentiles; robustness to alternative winsorization is in the Appendix.

3.2. Sample Construction

The raw dataset comprises 479,330 POI observations across 33,144 distinct items, 51,059 purchase orders, and 100 PBUs (descriptive statistics in Online Appendix, Table A.1).

The analysis sample is restricted to items for which at least one ordinary and at least one litigated purchase is observed, ensuring within-item comparability. This yields 226,305 observations covering 3,856 items.[2] For regressions using bid prices, we further restrict to winning bids (196,995 observations).

Several features of the data are noteworthy. In levels, ordinary purchases have substantially higher mean reference prices (R$909) compared to admin-

references that are present or absent independently of the price outcomes we estimate, so any residual misclassification attenuates rather than biases our estimates.

[2]Sample sizes vary slightly across tables due to missing values in specific dependent variables and the restriction to winners for price regressions.

istrative (R$226) and litigated (R$370) purchases, reflecting the mix of items across procurement types. However, the composition effect is absorbed by our item fixed effects. In logs—which form the basis of our regressions—litigated purchases show higher mean prices (1.56 for reference price, 1.02 for negotiated price) than ordinary purchases (0.93 and 0.32), consistent with a price premium conditional on the item. Quantities are markedly lower for litigated purchases (mean of 9,646 units vs. 31,487 for ordinary), reflecting the inability to aggregate demand under compressed timelines. The number of bidding firms is also lower for urgent purchases (2.2 for litigated vs. 3.2 for ordinary), suggesting reduced competition in the external phase.

The Online Appendix presents complementary geographic evidence on the spatial distribution of administrative demand, which mirrors the patterns documented for litigation in Section 2.

For the "under the gun" analysis, we construct a subsample of urgent purchases (administrative and litigated) for items with both types present, yielding 56,803 observations (balance statistics in Online Appendix, Table A.2). The tightest specifications use item×year-month cells: of 40,966 such cells, 4,402 (11%) carry within-cell variation in regime; the remaining cells are absorbed as singletons under the fixed-effects design. Both-types cells are not a representative subsample of urgent items—they tilt toward higher-volume, more-reordered medications. Online Appendix Table A.20 compares both-types cells with singletons along reference price, modality, SUS-formulary status, and supplier base, and we discuss how this representativeness restriction interacts with the bounded UTG estimate in Section 4.

4. Empirical Strategy

The analysis has two stages. We first estimate the urgent-vs-ordinary cost margin under item, time, and buyer fixed effects. We then bound the under-the-gun (UTG) gap between litigated and administrative urgent purchases under selection into the administrative channel. The identification of the within-item gap is straightforward; the identification of the UTG gap requires explicit selection bounds.

4.1. Identification: urgent vs. ordinary

Identification of the urgent-vs-ordinary premium requires that, conditional on item, time, and PBU fixed effects, urgency is uncorrelated with unobserved determinants of procurement outcomes. Three institutional features support this. First, court orders and administrative requests originate from patients' health needs and legal or scientific-committee assessments—not from the procurement process. Second, urgent demand is poorly correlated with PBU-level mismanagement: only 4% of claims arise from stock shortages or service failures. Third, the urgency regime is determined externally (by judges or the SES/SP scientific committee), not by the purchasing unit. We frame this design as a within-item descriptive estimate of the cost margin associated with the urgent regime.

A Borusyak-Jaravel-Spiess (2024) imputation event study around each item's first court order provides a dynamic check (Online Appendix Figure A.15). Pre-period coefficients are economically small (max absolute pre-period coefficient = 0.041); post-period coefficients rise from 0.052 at $t = 0$ to 0.147 at $t = +5$. We honest-test this dynamic pattern using Rambachan-Roth (2023) sensitivity: the breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero is $M = 0.018$. The $t = 0$ effect is significantly positive at zero pre-period violation but does not survive a linear extrapolation of the observed pre-period maximum. We honor this by treating the cross-sectional fixed-effects estimate as the primary object and the BJS event-study as supportive dynamic evidence rather than a stand-alone identification claim.

4.2. Identification: under-the-gun

Both litigated and administrative urgent purchases face identical planning constraints—compressed timelines, small quantities, diverted budget resources, and the same BEC platform and bidding procedures. They differ only in penalty exposure. The within-item UTG comparison isolates the cost margin associated with the sanction regime, conditional on a selection wedge that we now bound formally.

Selection into the administrative channel.. Cases enter the administrative channel rather than the courts when the patient (or a SUS social worker) requests the medication directly through an SES/SP intake and the request is approved by a scientific committee using cost-effectiveness criteria. The committee admits clinically essential and sourceable cases; rejects expensive, specialized, or uncertain ones. We do not treat the resulting selection bias as "ambiguous in sign"—the committee's first-stage probit on pre-period item characteristics rejects a higher pre-period log reference price (-0.051, $p < 0.001$) and SUS-formulary status (-0.265, $p < 0.001$): admin systematically over-represents items that would have been priced lower under *any* regime. The naive UTG comparison therefore inflates the sanction effect.

Manski-Lee bounds.. We bound the UTG coefficient under monotone selection by trimming admin observations within item×year×PBU strata where the admin sample exceeds the litigated count. The trimming proportion $p_{\text{trim},s} = \max\{0, (n_{\text{adm},s} - n_{\text{lit},s})/n_{\text{adm},s}\}$ is the share of admin observations within stratum s that, under the Lee monotone-acceptance assumption, would not have appeared in the litigated counterfactual. Trimming admin from the upper tail of the within-stratum price distribution yields the lower bound on the admin-minus-lit log-price gap; trimming from the lower tail yields the upper bound. Average trimming intensity is 26.9% of admin observations within strata where trimming applies; 12,038 of 44,654 strata trigger trimming. Section 5.2 reports the bounded UTG.

Heckman parametric correction (sensitivity).. A parametric Heckman correction with the same first-stage probit and item, year, and PBU FE in the second stage produces a second-stage Admin coefficient of 0.817 (SE 4.278, sample size 5,351). The inverse Mills ratio is collinear with the second-stage fixed effects; the resulting point estimate is uninformative. We treat the

Heckman as a sensitivity diagnostic, not as a competing point estimate, and rely on the Lee bound as the primary selection-corrected object.

Inference.. Standard errors are clustered at the PBU level. With approximately 99 clusters, asymptotic CR-cluster SEs may be liberal in the tightest specifications. We re-do inference for the UTG coefficients under a Rademacher wild cluster bootstrap ($B = 999$). The bootstrap p-value on the preferred-FE Admin coefficient is 0.0080; the corresponding 95% CI in log-points is $[-0.469, -0.039]$. The naive-UTG result survives wild-cluster inference; the within-item×year-month specification is borderline ($p = 0.0390$), as expected given the thin within-cell sample documented below.

Within-cell representativeness.. Item fixed effects absorb time-invariant item characteristics; item × year-month FE absorb time-varying within-item composition. The tightest specification therefore identifies off cells in which both administrative and litigated purchases occur in the same item×year-month. Of 40,966 such cells, 5,581 (about 11%) carry within-cell variation; the remaining 11,972 are absorbed as singletons under the FE design. Both-types cells are not a representative subsample (Online Appendix Table A.20): they tilt toward higher-volume, lower-priced, more-frequent SUS-formulary items than singleton cells. The bounded UTG estimate therefore speaks to the more liquid items in the urgent panel, not to the universe of court-mandated procurement. We discuss this restriction explicitly in the welfare bound calculation in Section 5.

4.3. Specifications

We estimate the urgent-vs-ordinary premium as

$$y_{i,g,t} = \beta \cdot \text{Urgent}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{1}$$

where $y_{i,g,t}$ is the outcome (log negotiated price, log reference price, log number of firms, success indicator) for purchase order i of item g at time t; γ_g, λ_t, δ_b are item, time, and PBU fixed effects. We report four specifications with progressively richer fixed effects.

We estimate the under-the-gun coefficient as

$$\ln(\text{Price}_{i,g,t}) = \beta \cdot \text{Admin}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{2}$$

on the urgent-only subsample (admin or litigated). We report (i) the naive coefficient under the preferred FE; (ii) the Manski-Lee bounded coefficient; (iii) the within firm-buyer-item triple coefficient that isolates same-firm pricing from supplier composition.

We do not estimate a structural counterfactual price under hypothetical sanction removal. That object is unidentified in our setting because the administrative subsample is selected on item economics. What we provide is a bounded descriptive decomposition of where the cost margin between regimes comes from.

5. Results

We organize the results around the central conceptual contrast of the paper: the urgency premium operates through *sourcing* (which firm wins, what order size is feasible) and not through within-firm *pricing*. We first document the headline urgent-vs-ordinary patterns, then bound the under-the-gun (UTG) gap under selection, then present the within firm-buyer-item null that pivots the interpretation, and finally reconcile the bounded UTG with its mechanical components.

5.1. Urgency raises prices and thins competition; tenders still clear

Urgent purchases carry negotiated prices 5.4% higher than ordinary purchases of the same item, with the same item attracting -5.4% fewer bidding firms. Reference prices set during the planning phase rise by 2.7% in the preferred specification ($p < 0.10$); the raw item-FE-only premium is 17.8%, attenuating sharply once buyer fixed effects absorb buyer-level heterogeneity. Negotiated prices show the same pattern, ranging from 17.3% with item FE alone to 5.4–5.7% with our preferred specifications.

Despite worse conditions, urgent tenders *succeed* more often: 2.1 pp ($p < 0.01$). This rules out the alternative that urgent purchases are simply cancelled when the bid distribution is unfavourable; officials accept worse terms to ensure compliance. The contrast with the cross-buyer dispersion benchmarks in the literature (Bandiera-Prat-Valletti 2009: 10–20%; Best-Hjort-Szakonyi 2023: 20–40%; Bosio et al. 2022: 15–25%; Coviello-Guglielmo-

Spagnolo 2018: 5–12%) places our 5.4% within-item premium at the lower end of plausible passive-waste magnitudes; the same-buyer, same-item contrast is tighter than typical cross-buyer dispersion, so a smaller headline number is consistent with the design.

Table 2: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	0.159***	0.151***	0.053***	0.056***
	(0.042)	(0.038)	(0.016)	(0.016)
Panel B: Direct Effect				
Urgent Purchase	0.075	0.060	0.030*	0.030*
	(0.045)	(0.042)	(0.017)	(0.017)
Log Quantity	-0.321***	-0.325***	-0.392***	-0.393***
	(0.026)	(0.026)	(0.029)	(0.029)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883
Within R^2 (A)	0.003	0.003	0.000	0.000
Within R^2 (B)	0.315	0.327	0.358	0.359

Notes: Dependent variable: log negotiated price. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

5.2. The under-the-gun comparison and selection bounds

The naive UTG comparison contrasts litigated and administrative urgent purchases of the same item. Both share planning constraints; they differ only in penalty exposure. The naive coefficient on *Admin* is -0.262

Table 3: Manski-Lee bounds on the under-the-gun coefficient.

	log negotiated price		implied
Specification	Coef. on Admin	SE	Lit-vs-Admin (%)
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530
Lee lower bound (admin top-trim)	-0.192	0.095	15.924
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123

Trimming applied within item*times*year*times*PBU strata where the admin sample exceeds the li

($p < 0.05$) under the preferred specification, implying litigated purchases are 29.5% more expensive than administrative ones. Wild cluster bootstrap inference (Rademacher weights, $B = 999$) confirms the result with $p = 0.0080$ and a 95% CI in log-points of $[-0.469, -0.039]$.

This comparison is contaminated by selection into the administrative channel: the SES/SP scientific committee admits cases on cost-effectiveness criteria. The committee's selection rule, recovered from a probit on pre-period item characteristics, is informative ($N = 5{,}351$): items with higher pre-period reference prices are less likely to be admitted (probit coefficient -0.051, $p < 0.001$); SUS-formulary items are less likely (probit coefficient -0.265, $p < 0.001$). Admin therefore over-represents items that would have been priced lower under *any* regime.

We bound the selection wedge using Lee (2009) trimming within item×year×PBU strata where the admin sample exceeds the litigated count. Trimming admin observations from the top of the within-stratum price distribution gives the lower bound; trimming from the bottom gives the upper bound. Table A.18 reports the bounded UTG.

The bounded UTG falls in $[15.9\%, 21.1\%]$, well below the naive 29.5%. Selection on observables explains approximately one-third of the headline gap.[3] The within-cell variation that identifies the tightest specification comes from a non-representative subsample (Online Appendix Table A.20): of 40,966 item×month cells, only 5,581 (about 11%) carry within-cell variation and they tilt toward higher-volume, more-frequent SUS-formulary medications. Bounded UTG estimates therefore speak to the lower-volume liquid items in the sample, not to the universe of court-mandated purchases.

5.3. Same firm, same buyer, same item: no within-firm markup

The interpretation of the bounded UTG hinges on *how* the gap is generated. We separate within-firm pricing from equilibrium supplier selection by restricting to firm-buyer-item triples observed under both the litigated and the administrative regime within urgent procurement. There are 1,206 such triples covering 4,573 observations.

The within firm-buyer-item Admin coefficient is 0.035 (SE 0.041), *not significantly different from zero.* The same firm sells the same item to the same buyer at essentially the same per-unit price under both regimes. There is no within-firm markup under sanctions. Whatever distinguishes litigated from administrative urgent procurement, it is not opportunistic pricing by

[3]We also report a Heckman parametric correction with the same first-stage probit (Online Appendix Table A.19). The Mills ratio is collinear with the second-stage fixed effects; the resulting Admin coefficient (0.817, SE 4.278) is uninformative. We treat the Lee bound as the primary selection-corrected object and the Heckman estimate as a sensitivity diagnostic that flags identification fragility, not as a competing point estimate.

suppliers that already serve the buyer's procurement.

This finding is the empirical pivot of the paper. In the standard accountability-distortion intuition (Prendergast, 2007), sanction-exposed officials would face higher per-unit prices because suppliers extract a premium for delivery risk. In our setting, that mechanism is empirically absent.

5.4. Reconciling the bounded UTG: mechanical fragmentation, sourcing residual

We decompose the observed log-price gap (admin minus litigated) into three components linked by an arithmetic identity:

$$\underbrace{\Delta y_{\text{obs}}}_{-0.259} = \underbrace{\beta_{\text{bulk}} \cdot \Delta \ln Q_{\text{adm-lit}}}_{\text{mechanical C1: } -0.397} + \underbrace{\beta_{\text{within-firm}}}_{\text{within-firm: } +0.035} + \underbrace{\beta_{\text{composition}}}_{\text{residual: } +0.103} .$$

Table A.21 reports the three components and their implied per-unit-price magnitudes.

Three readings follow. *First*, the mechanical bulk-discount channel (−32.8%, admin cheaper by 32.8% from order-size differences alone) over-explains the

Table 4: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log-points (admin minus lit)
Observed UTG (admin minus lit, log price)	-0.259
Mechanical C1: bulk-discount × qty gap	-0.397
Within firm-buyer-item offset	0.035
Supplier composition residual	0.104

Identity: observed UTG = mechanical C1 + within-firm offset + composition residual. Within-fi observed gap. The qty channel pulls the admin-vs-lit price gap below what we observe.

Second, the within firm-buyer-item offset is small and not significant (3.5%, NULL). Within-firm pricing behavior is not where the residual wedge lives.

Third, the supplier composition residual is positive (10.9%): holding quantity constant and within firms not observed in both regimes, the litigated supplier mix is *cheaper* by about 11% than the administrative supplier mix. This reverses the sign one might expect under naive intuition: the sanction-exposed regime forces sourcing onto a different (and on average less expensive per unit) point of the supplier distribution, but the bulk-discount disadvantage of small lit orders dominates, producing the net 23% lit-over-admin observed gap. The cost of sanctions, in this setting, is a sourcing-and-aggregation cost, not a pricing cost.

Figure 1 visualizes the decomposition.

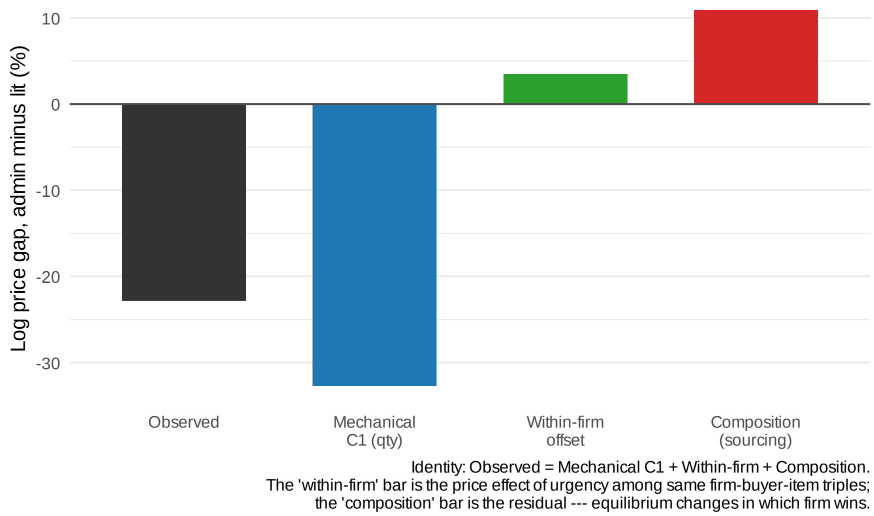


Figure 1: Sourcing vs. pricing: reconciliation of the under-the-gun gap.
Notes: Bars give the four components of the identity in Table A.21, expressed as percent (admin minus lit). The within firm-buyer-item offset is not statistically distinguishable from zero. The composition residual is the part of the observed UTG that operates through equilibrium changes in *which* firm wins the contract.

5.5. Falsification: items never litigated

If the urgency premium reflects judicial pressure rather than unobserved item characteristics, it should be absent for items that are never subject to court orders. Running the preferred specification on items with zero litigated purchases across 2009–2019 (restricted to those with both ordinary and administrative purchases) yields placebo coefficients of -0.020 (SE 0.032) for negotiated prices and -0.031 (SE 0.045) for reference prices. Both are economically small and statistically insignificant (Online Appendix Table A.8). The urgency premium exists only for items that are targets of litigation.

5.6. *Where the cost bites hardest*

Splitting items by mean number of participating firms (median 2.071), competitive items show a 14.4% urgency premium with 8.5% surviving firm fixed effects (39% attenuation). Concentrated items show no detectable premium (-1.0%, not significant). The pattern matches the sourcing reading: urgency erodes competitive pressure where there is competitive pressure to erode. Where the buyer has no outside option in either regime, urgency carries little additional cost.

5.7. *Summary of effects*

Across the full sample, the 5.4% urgent-vs-ordinary premium reflects compressed planning timelines and reduced bidder participation. Within urgent procurement, the bounded UTG falls in $[15.9\%, 21.1\%]$ once selection into the administrative channel is corrected. The bounded UTG itself decomposes into a mechanical bulk-discount fragmentation channel, a null within-firm pricing component, and a positive supplier composition residual. Same firms price the same items the same way regardless of regime. Different firms win urgent contracts under the two regimes, and the sourcing shift is the empirical signature of accountability-induced inefficiency in our data.

6. Conclusion

Courts that compel governments to buy medications save individual lives. They also raise public procurement costs by 5.4% per purchase across the full sample, attract -5.4% fewer bidders, and—once selection into the SES/SP administrative channel is bounded—generate an under-the-gun gap of $[15.9\%, 21.1\%]$ between litigated and administrative urgent purchases of the same item.

The empirical signature of accountability-induced inefficiency in our data is *sourcing*, not *pricing*. Within firm-buyer-item triples observed under both the litigated and the administrative regime, the same firm sells the same item to the same buyer at essentially the same per-unit price (3.5%, NULL). Suppliers that already serve the buyer's urgent procurement do not extract a markup when the official is sanction-exposed. The cost margin that does exist operates through demand fragmentation—compressed timelines force smaller orders, mechanically losing bulk discounts—and through equilibrium shifts in which firm wins each contract. The reconciliation in Table A.21 makes this explicit: of the observed gap, the mechanical bulk-discount channel over-explains the magnitude; the within-firm component is zero; the supplier composition residual offsets the bulk-discount overshoot to deliver the net observed lit-over-admin gap.

Welfare.. Applied to the \$300 M of annual litigated spending in São Paulo (CNJ/INSPER, 2019) and a calibration anchor of 50.0% of cases admissible to the administrative channel under expanded committee capacity, the bounded UTG implies a per-unit-price welfare bound of \$27.8 M per year, with the Lee-endpoint range delivering $[\$23.9\ M, \$31.7\ M]$. This is a per-unit-price margin and excludes welfare losses from constrained order sizes (the mechanical bulk-discount channel itself), reduced bidder participation, and officials' search time diverted from ordinary procurement. We report a single defensible figure rather than the stacked ranges common in the policy literature.

Policy implications.. Two reforms address the two channels we identify, with first-order costs the policy maker can plausibly bound.

On the demand side, *expanding the administrative-request mechanism* eliminates within-urgent sanction exposure for the admissible share. Since the within-firm pricing component is zero, the gain from this reform is realized through the demand-aggregation margin: admin orders are larger, so the bulk-discount channel works in the buyer's favor. The instrument is administrative capacity (the scientific-committee infrastructure), not legal reform; the admissibility share is the binding parameter. We do not endorse a specific calibration; we show that under any plausible value the welfare gain scales with bounded UTG times \$300 M.

On the supply side, *framework agreements for repeat-purchase commodity-like medications* address demand fragmentation directly by allowing aggregation across urgent and ordinary requests. The instrument is contract-design reform, available under Lei 14.133/2021. We do not attach a recovery point estimate to this channel: any number requires a calibration anchor on the share of items eligible for framework agreements, and we are not in a position to provide one without further institutional input.

Interpretation.. The findings highlight a tension at the heart of right-to-health enforcement. Judicial mandates intended to guarantee individual access impose substantial procurement costs on the state, but—in our data—the mechanism is not a markup story. It is an aggregation and sourcing story. Institutional designs that preserve the individual right while restoring the demand-aggregation and supplier-search margins compromised under compressed timelines—through framework agreements, expanded administrative-channel capacity, or both—can deliver welfare gains for litigants and the broader population simultaneously. The recent Brazilian procurement reform (Lei 14.133/2021), which expands electronic auctions and introduces framework agreements, sets up a natural test of this prediction; future work using post-reform data can quantify the realized recovery.

Boundary.. Our results are estimated within São Paulo's BEC platform, on pharmaceutical procurement in a single state, in 2009-2019. We do not extrapolate to other states, to non-pharmaceutical procurement, or to post-reform institutional arrangements. The Lee bounds apply within the urgent panel; the welfare figure assumes that admin admissibility can be expanded without affecting the per-unit-price gap, an assumption the manuscript does not test.

References

Acharya, A., Blackwell, M., Sen, M., 2016. Explaining causal findings without bias: Detecting and assessing direct effects. American Political Science

Review 110, 512–529. doi:10.1017/S0003055416000216.

Bandiera, O., Best, M.C., Khan, A.Q., Prat, A., 2021. The allocation of authority in organizations: A field experiment with bureaucrats. Quarterly Journal of Economics 136, 2195–2242. doi:10.1093/qje/qjab029.

Bandiera, O., Prat, A., Valletti, T., 2009. Active and passive waste in government spending: Evidence from a policy experiment. American Economic Review 99, 1278–1308. doi:10.1257/aer.99.4.1278.

Best, M.C., Hjort, J., Szakonyi, D., 2023. Individuals and organizations as sources of state effectiveness. American Economic Review 113, 2121–2167. doi:10.1257/aer.20191598.

Biehl, J.a., Petryna, A., Gertner, A., Amon, J.J., Picon, P.D., 2009. Judicialisation of the right to health in Brazil. The Lancet 373, 2182–2184. doi:10.1016/S0140-6736(09)61172-7.

Borusyak, K., Jaravel, X., Spiess, J., 2024. Revisiting event-study designs: Robust and efficient estimation. Review of Economic Studies 91, 3253–3285. doi:10.1093/restud/rdae007.

Bosio, E., Djankov, S., Glaeser, E.L., Shleifer, A., 2022. Public procurement in law and practice. American Economic Review 112, 1091–1117. doi:10.1257/aer.20200738.

Callaway, B., Sant'Anna, P.H.C., 2021. Difference-in-differences with multiple time periods. Journal of Econometrics 225, 200–230. doi:10.1016/j.jeconom.2020.12.001.

Carril, R., Gonzalez-Lira, A., Walker, M.S., 2026. Competition under incomplete contracts and the design of procurement policies. American Economic Review 116, 535–581. doi:10.1257/aer.20221345.

Castro, M.C., Massuda, A., Almeida, G., Menezes-Filho, N.A., Andrade, M.V., de Souza Noronha, K.V.M., Rocha, R., Macinko, J., Hone, T., Tasca, R., Giovanella, L., Malik, A.M., Werneck, H., Fachini, L.A., Atun, R., 2019. Brazil's unified health system: The first 30 years and prospects for the future. The Lancet 394, 345–356. doi:10.1016/S0140-6736(19)31243-7.

CNJ/INSPER, 2019. Judicialização da saúde no Brasil: Perfil das demandas, causas e propostas de solução. Conselho Nacional de Justiça, Brasília.

Coviello, D., Moretti, L., Spagnolo, G., Valbonesi, P., 2018. Court efficiency and procurement performance. Scandinavian Journal of Economics 120, 826–858. doi:10.1111/sjoe.12225.

Decarolis, F., Giuffrida, L.M., Iossa, E., Mollisi, V., Spagnolo, G., 2020. Bureaucratic competence and procurement outcomes. Journal of Law, Economics, and Organization 36, 537–597. doi:10.1093/jleo/ewaa004.

Ferraz, O.L.M., 2009. The right to health in the courts of Brazil: Worsening health inequities? Health and Human Rights 11, 33–45.

Prendergast, C., 2007. The motivation and bias of bureaucrats. American Economic Review 97, 180–196. doi:10.1257/aer.97.1.180.

Rambachan, A., Roth, J., 2023. A more credible approach to parallel trends. Review of Economic Studies 90, 2555–2591. doi:10.1093/restud/rdad018.

Rasul, I., Rogger, D., 2018. Management of bureaucrats and public service delivery: Evidence from the Nigerian civil service. Economic Journal 128, 413–446. doi:10.1111/ecoj.12418.

Soares, A., 2019. Health system financing paradigm in the state of São Paulo: A regional analysis. Revista de Saúde Pública 53, 39. doi:10.11606/S1518-8787.2019053000796.

Wang, D.W.L., 2015. Right to health litigation in Brazil: The problem and the institutional responses. Human Rights Law Review 15, 617–641. doi:10.1093/hrlr/ngv025.

Williams, M.J., 2021. Beyond state capacity: Bureaucratic performance, policy implementation, and reform. Journal of Institutional Economics 17, 339–357. doi:10.1017/S1744137420000478.

Online Appendix

This appendix provides tables and figures from the main text, robustness checks, and supplementary evidence.

A.0 Supporting Descriptives, Balance, and Secondary Outcome Tables

Table A.1: Descriptive Statistics by Purchase Type

	Ordinary		Administrative		Litigated		Diff	Diff
	Mean	(SD)	Mean	(SD)	Mean	(SD)	(O–L)	(A–L)
Panel A: Levels								
Reference Price	909.17	(5,959.41)	226.38	(2,257.35)	370.39	(2,540.20)	538.78***	-144.01***
							[0.000]	[0.000]
Negotiated Price	477.51	(2,915.88)	148.03	(1,177.52)	260.25	(1,561.81)	217.26***	-112.22***
							[0.000]	[0.000]
Quantity	31,487.15	(163,507.29)	117,146.00	(356,962.37)	9,646.19	(82,683.10)	21,840.96***	107,499.81***
							[0.000]	[0.000]
N. Bidding Firms	3.17	(1.93)	2.43	(1.35)	2.20	(1.19)	0.97***	0.23***
							[0.000]	[0.000]
Panel B: Log Transformations								
Log Reference Price	0.926	(2.723)	0.909	(2.541)	1.557	(2.446)	-0.632***	-0.648***
							[0.000]	[0.000]
Log Negotiated Price	0.319	(2.862)	0.480	(2.604)	1.024	(2.568)	-0.705***	-0.544***
							[0.000]	[0.000]
Log Quantity	6.947	(2.678)	7.472	(3.160)	6.087	(2.083)	0.860***	1.385***
							[0.000]	[0.000]
Log N. Firms	0.984	(0.590)	0.747	(0.529)	0.665	(0.493)	0.319***	0.081***
							[0.000]	[0.000]
Panel C: Tender Characteristics								
Successful Tender (%)	0.859	(0.348)	0.869	(0.338)	0.908	(0.288)	-0.050***	-0.040***
							[0.000]	[0.000]
Observations	136,250		15,371		45,351			

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch t-tests; p-values in brackets. *** p<0.01, ** p<0.05, * p<0.1.

Section A.1 examines the sensitivity of the main estimates to the choice of winsorization level. Section A.2 assesses the robustness of the "under the gun" estimates to progressive inclusion of control variables. Section A.3 presents distributional evidence on key outcome variables and additional descriptive figures. Section A.4 presents geographic descriptive evidence on

Table A.2: Balance Table: Administrative vs Litigated (Urgent Purchases)

	Administrative		Litigated		Diff	*p*-value
	Mean	(SD)	Mean	(SD)	(A–L)	
Panel A: Procurement Outcomes						
Reference Price	226.38	(2,257.35)	232.56	(1,432.18)	-6.18	[0.752]
Negotiated Price	148.03	(1,177.52)	174.79	(1,052.56)	-26.75**	[0.013]
Quantity	117,146.00	(356,962.37)	9,306.87	(80,252.25)	107,839.13***	[0.000]
Log Reference Price	0.91	(2.54)	1.43	(2.32)	-0.52***	[0.000]
Log Negotiated Price	0.48	(2.60)	0.89	(2.44)	-0.41***	[0.000]
Log Quantity	7.47	(3.16)	6.18	(1.99)	1.29***	[0.000]
Panel B: Market Structure						
N. Bidding Firms	2.427	(1.351)	2.155	(1.120)	0.272***	[0.000]
Log N. Firms	0.747	(0.529)	0.650	(0.483)	0.097***	[0.000]
Successful Tender (%)	0.869	(0.338)	0.914	(0.280)	-0.045***	[0.000]
Panel C: Purchase Characteristics						
Electronic Auction	0.985	(0.123)	0.941	(0.236)	0.044***	[0.000]
Observations	15,371		41,491			

Notes: Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%. *p*-values from Welch *t*-tests in brackets. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

administrative demand, complementing the litigation maps shown in Section 2. Section A.5 presents an event study around the first court order for each item. Section A.6 examines heterogeneity in the urgency premium along several dimensions.

Table A.3: Within-Item Balance: Administrative vs. Litigated

Covariate	Mean (Admin)	Mean (Lit)	Raw diff. (A–L)	Within-item diff. Coef	(SE)
SUS basic (MEDICAMENTO flag)	0.718	0.892	-0.174	0.000^{NA}	(NaN)
Electronic auction (pregão)	0.984	0.941	0.043	0.041^{**}	(0.017)
Log quantity (bulk size signal)	7.642	6.180	1.462	0.866^{*}	(0.502)
Log reference price	0.835	1.401	-0.566	-0.339^{***}	(0.093)
Successful tender	0.869	0.914	-0.045	-0.024	(0.022)
Late period (year $\geq$ 2014)	0.565	0.623	-0.058	-0.045	(0.041)
Large PBU (above-median)	0.895	0.810	0.085	0.083	(0.073)
Observations			63,426		
Items			2,370		

Notes: Within-item balance between administrative ($Admin = 1$) and litigated ($Admin = 0$) urgent purchases, restricted to items with both types present. Raw difference is the unconditional mean gap. Within-item coefficient is β from $x_{i,g,t} = \beta\,\mathrm{Admin}_{i,g,t} + \gamma_g + \varepsilon_{i,g,t}$, where γ_g is an item fixed effect and standard errors are clustered at the PBU level. A coefficient close to zero means that, within a given item, the administrative and litigated draws are balanced on the covariate. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A.4: Reference Prices

	(1)	(2)	(3)	(4)
Urgent Purchase	0.164***	0.156***	0.027*	0.027*
	(0.056)	(0.050)	(0.014)	(0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R^2	0.003	0.003	0.000	0.000

Notes: Dependent variable: log reference price. Sample: winners only, items with both ordinary and litigated purchases. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table A.5: Quantities

	(1)	(2)	(3)	(4)
Urgent Purchase	-0.264	-0.279	-0.058	-0.065
	(0.183)	(0.181)	(0.056)	(0.058)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R^2	0.003	0.003	0.000	0.000

Notes: Dependent variable: log quantity. Sample: winners only. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

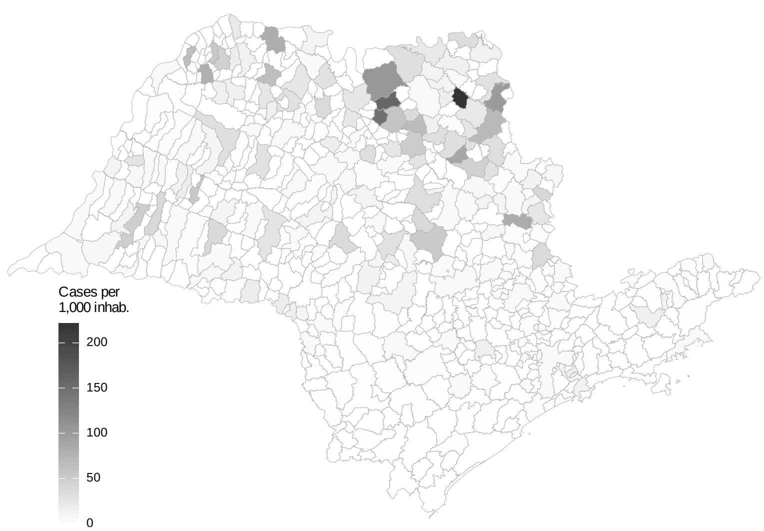


Figure A.1: Health Litigation Cases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors' elaboration based on CNJ/INSPER (2019) litigation data and IBGE population estimates.

Table A.6: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.014)	(0.015)
Panel B: Direct Effect				
Urgent Purchase	-0.091***	-0.102***	-0.042***	-0.040***
	(0.014)	(0.011)	(0.012)	(0.013)
Log Quantity	0.063***	0.064***	0.065***	0.065***
	(0.004)	(0.004)	(0.003)	(0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	81,805	81,805	81,801	81,801
Obs. (Panel B)	81,793	81,793	81,789	81,789
Within R^2 (A)	0.004	0.006	0.001	0.001
Within R^2 (B)	0.063	0.069	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.7: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel B: Direct Effect				
Urgent Purchase	0.023**	0.023**	0.021***	0.022***
	(0.009)	(0.010)	(0.006)	(0.006)
Log Quantity	0.003	0.003	0.012***	0.012***
	(0.004)	(0.004)	(0.002)	(0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	226,305	226,305	226,301	226,301
Obs. (Panel B)	226,282	226,282	226,278	226,278
Within R^2 (A)	0.001	0.001	0.000	0.000
Within R^2 (B)	0.001	0.001	0.004	0.004

Notes: Dependent variable: successful tender (LPM). Sample: all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.8: Placebo Test: Items Never Subject to Litigation

	bid_price_log		bid_price_ref_log	
	(1)	(2)	(3)	(4)
Urgent Purchase	-0.0204	0.0513***	-0.0306	0.0305**
	(0.0316)	(0.0153)	(0.0447)	(0.0141)
Observations	39,283	196,883	46,874	226,278
R^2	0.83364	0.86796	0.81121	0.85183
Within R^2	7.03×10^{-6}	0.00022	1.73×10^{-5}	7.43×10^{-5}
item_id fixed effects	✓	✓	✓	✓
year_n fixed effects	✓	✓	✓	✓
pbu_id fixed effects	✓	✓	✓	✓

Placebo sample: items with zero litigated purchases across 2009-2019. Main sample: items with at least one litigated and one ordinary purchase. DV: log price. Item + Year + PBU FE. SE clustered at PBU level.

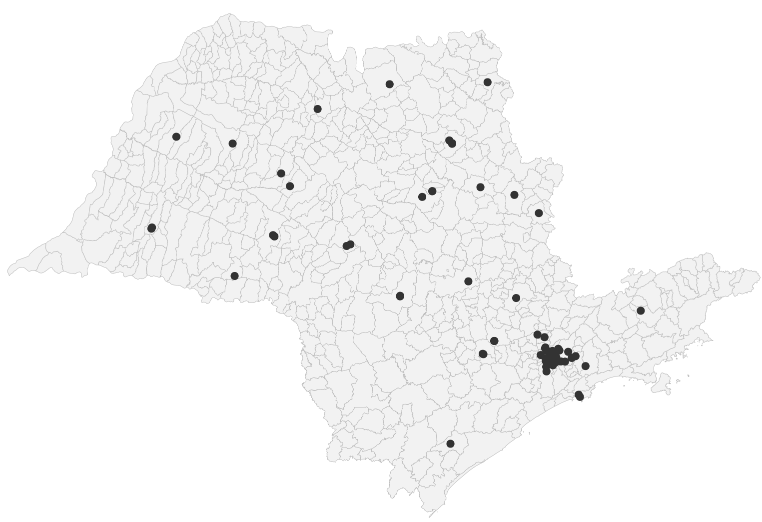


Figure A.2: Public Buyer Units (SES/SP)

Source: Authors' elaboration based on SES/SP administrative records.

A.1 Winsorization Sensitivity

Our baseline estimates use 1%/99% winsorization to limit the influence of extreme outliers while preserving meaningful variation in the data. To verify that our results are not driven by this specific choice, Tables A.9–A.12 replicate the main regression tables under three alternative winsorization treatments: no winsorization (Panel A), the 1%/99% baseline (Panel B), and a more aggressive 5%/95% winsorization (Panel C).

Table A.9 reports the sensitivity of the reference price estimates. The positive coefficient on urgent purchases is stable across all three winsorization levels: the preferred specification (column 3) yields estimates of 0.026 (no winsorization), 0.027 (baseline), and 0.027 with aggressive winsorization. The qualitative pattern—a large raw premium that attenuates substantially with PBU fixed effects—is preserved throughout, indicating that the reference price result is not driven by outliers.

Table A.10 presents the corresponding sensitivity analysis for negotiated prices. The total effect of urgency on negotiated prices is positive and statistically significant across all winsorization levels. The coefficient magnitudes are slightly larger without winsorization (reflecting the influence of extreme values) and slightly smaller with aggressive winsorization, but the economic message remains unchanged: urgent purchases are associated with higher negotiated prices, with the preferred estimate ranging from approximately 5% to 6%.

Table A.11 examines the robustness of the bidder participation results.

Table A.9: Robustness: Reference Prices

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.167***	0.159***	0.026*	0.026*
	(0.056)	(0.051)	(0.014)	(0.014)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.164***	0.156***	0.027*	0.027*
	(0.056)	(0.050)	(0.014)	(0.014)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.156***	0.150***	0.027*	0.028**
	(0.053)	(0.048)	(0.014)	(0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896

Notes: Dependent variable: log reference price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The negative effect of urgency on the number of bidding firms is remarkably stable across winsorization levels, with preferred estimates in the range of −0.06 to −0.05 log points. This stability is expected, since the number of firms is a count variable with limited extreme values, making it less sensitive to winsorization.

Table A.12 replicates the tender success analysis. The positive coefficient on urgency—indicating that urgent tenders are more likely to succeed—is robust across all winsorization levels. The magnitude is consistent at approximately 2 pp in the preferred specification, reinforcing the interpretation

Table A.10: Robustness: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.160***	0.152***	0.051***	0.054***
	(0.043)	(0.038)	(0.015)	(0.016)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.159***	0.151***	0.053***	0.056***
	(0.042)	(0.038)	(0.016)	(0.016)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.162***	0.153***	0.058***	0.061***
	(0.041)	(0.037)	(0.017)	(0.017)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883

Notes: Dependent variable: log negotiated price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

that procurement officials under pressure accept less favorable terms to ensure compliance.

Table A.11: Robustness: Participant Firms

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.015)	(0.015)
Panel B: Winsorized 1%/99%				
Urgent Purchase	-0.101***	-0.113***	-0.056***	-0.054***
	(0.023)	(0.022)	(0.014)	(0.015)
Panel C: Winsorized 5%/95%				
Urgent Purchase	-0.100***	-0.111***	-0.055***	-0.053***
	(0.022)	(0.021)	(0.014)	(0.015)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (A)	81,377	81,377	81,373	81,373
Obs. (B)	81,805	81,805	81,801	81,801
Obs. (C)	81,805	81,805	81,801	81,801

Notes: Dependent variable: log number of bidding firms. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.12: Robustness: Tender Success (LPM)

	(1)	(2)	(3)	(4)
Panel A: No Winsorization				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel B: Winsorized 1%/99%				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Panel C: Winsorized 5%/95%				
Urgent Purchase	0.023**	0.023**	0.021***	0.021***
	(0.009)	(0.009)	(0.006)	(0.006)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	226,305	226,305	226,301	226,301

Notes: Dependent variable: successful tender (LPM). Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

A.2 Under the Gun: Progressive Controls

Table A.13 examines the sensitivity of the "under the gun" estimate to the progressive inclusion of control variables and to the choice of winsorization level. Starting from the baseline specification with item + year + PBU fixed effects (column 1), we sequentially add log quantity (column 2), log reference price (column 3), log number of firms (column 4), and finally replace year FE with year-month FE (column 5). Each row within a panel shows how the coefficient on the administrative indicator responds to additional controls.

The key finding is that the administrative indicator remains negative across all specifications and winsorization levels, indicating that litigated purchases are consistently more expensive than administrative ones. The coefficient is largest and most precisely estimated in the baseline specification without additional controls (column 1), where it captures the total effect including indirect channels through quantity and competition. As controls are added, the coefficient attenuates—consistent with some of the price premium operating through the quantity and competition channels—but the sign is preserved throughout. This pattern confirms that the sanction channel has an economically meaningful direct effect on prices beyond its indirect effects through procurement conditions.

Table A.13: Robustness: Under the Gun with Progressive Controls

	(1)	(2)	(3)	(4)	(5)
Panel A: No Winsorization					
Administrative (vs Litigated)	-0.259**	0.138	0.115**	0.201**	0.195**
	(0.095)	(0.130)	(0.054)	(0.090)	(0.087)
Panel B: Winsorized 1%/99%					
Administrative (vs Litigated)	-0.262**	0.117	0.110*	0.199**	0.194**
	(0.096)	(0.122)	(0.055)	(0.093)	(0.090)
Panel C: Winsorized 5%/95%					
Administrative (vs Litigated)	-0.261**	0.060	0.089*	0.169*	0.165*
	(0.096)	(0.093)	(0.049)	(0.084)	(0.080)
Log Quantity	No	Yes	Yes	Yes	Yes
Log Ref. Price	No	No	Yes	Yes	Yes
Log N. Firms	No	No	No	Yes	Yes
Item FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No
Year-Month FE	No	No	No	No	Yes
PBU FE	Yes	Yes	Yes	Yes	Yes
Obs. (A)	56,803	56,803	56,803	14,913	14,913
Obs. (B)	56,803	56,803	56,803	15,052	15,052
Obs. (C)	56,803	56,803	56,803	15,052	15,052

Notes: Dependent variable: log negotiated price. UTG sample (urgent, winners, items with both administrative and litigated). Progressive controls added left to right. Column (5) replaces Year with Year-Month FE. Standard errors clustered at the PBU level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

A.3 Distributional Evidence

The regression estimates in the main text capture average treatment effects. The figures below complement these estimates by showing how the *distributions* of key outcome variables differ across purchase types, providing visual evidence of the systematic shifts that underlie our regression results.

Figure A.3 plots kernel density estimates of log reference prices separately for ordinary, administrative, and litigated purchases. The rightward shift of the litigated distribution relative to ordinary purchases is clearly visible, consistent with the positive coefficient on urgency in Table A.4. The administrative distribution lies between the other two, reflecting the intermediate planning conditions of this purchase type.

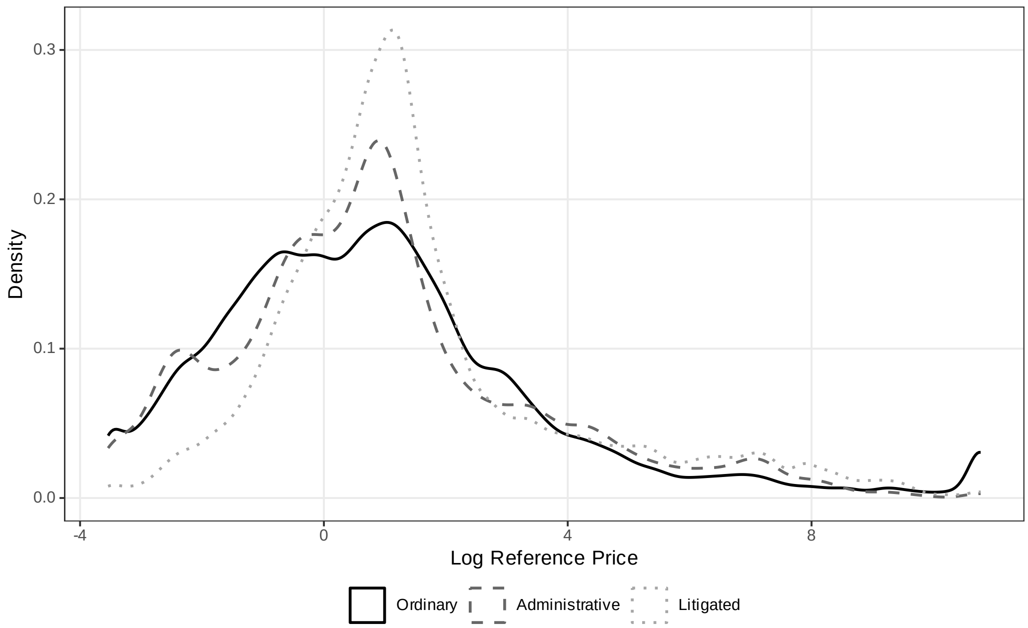


Figure A.3: Distribution of Log Reference Price by Purchase Type

Figure A.4 presents the analogous comparison for log negotiated prices. The distributional shift is qualitatively similar to reference prices, with litigated purchases showing higher negotiated prices on average. Notably, the litigated distribution also exhibits a somewhat thicker right tail, suggesting that a subset of litigated purchases involves particularly large price premiums.

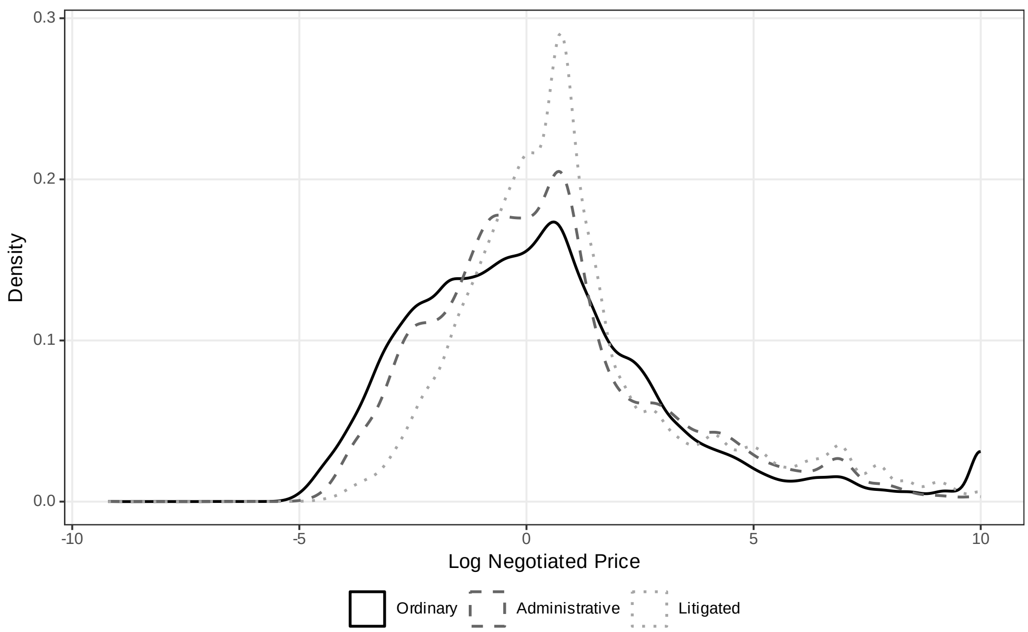


Figure A.4: Distribution of Log Negotiated Price by Purchase Type

Figure A.5 shows the distribution of log quantities. In contrast to prices, here the litigated distribution is shifted *leftward*, reflecting the smaller quantities demanded in urgent purchases. The ordinary distribution has a heavier right tail, consistent with the ability to aggregate demand under standard planning timelines. The administrative distribution is bimodal, spanning both small (individual prescription) and larger (stock replenishment) quantities.

Figure A.6 displays the distribution of log number of bidding firms. The litigated distribution is clearly shifted leftward relative to ordinary purchases, with a larger mass at low participation levels. This visual pattern confirms the regression finding that urgent tenders attract fewer bidding firms, reduc-

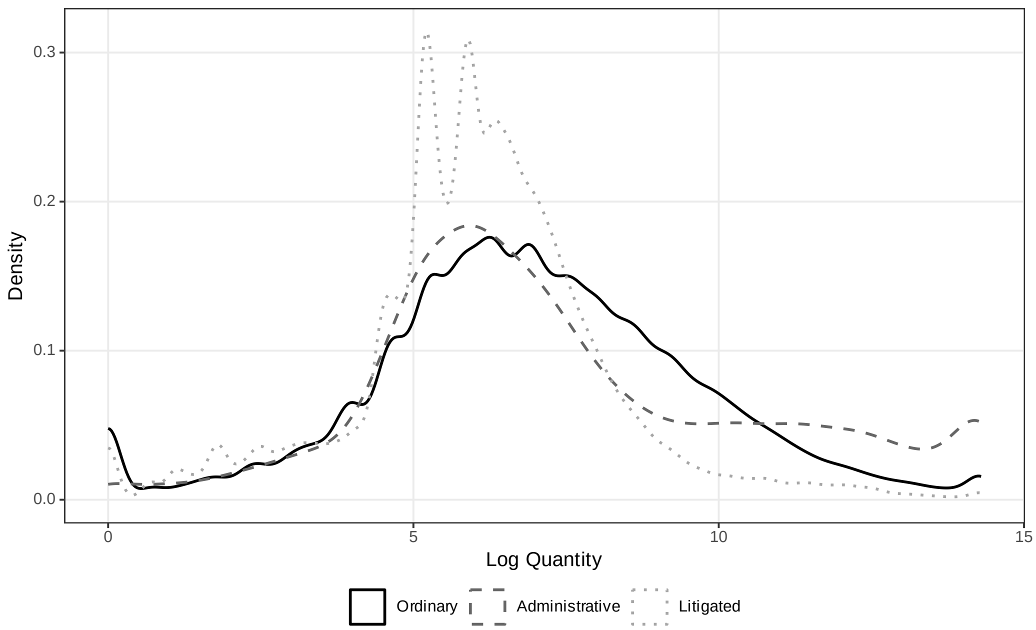


Figure A.5: Distribution of Log Quantity by Purchase Type

ing the competitive pressure that normally restrains prices.

Figure A.7 focuses on the "under the gun" subsample, comparing the distributions of log negotiated prices for administrative and litigated urgent purchases only. The rightward shift of the litigated distribution relative to administrative purchases provides direct visual evidence of the sanction channel: holding constant the urgency of the purchase, judicial pressure shifts the entire price distribution to the right.

Figure A.8 compares tender success rates across purchase types. Consistent with the LPM estimates, litigated purchases have the highest success rate, followed by administrative and ordinary purchases. This ordering is consistent with the "under the gun" mechanism: the greater the penalty for

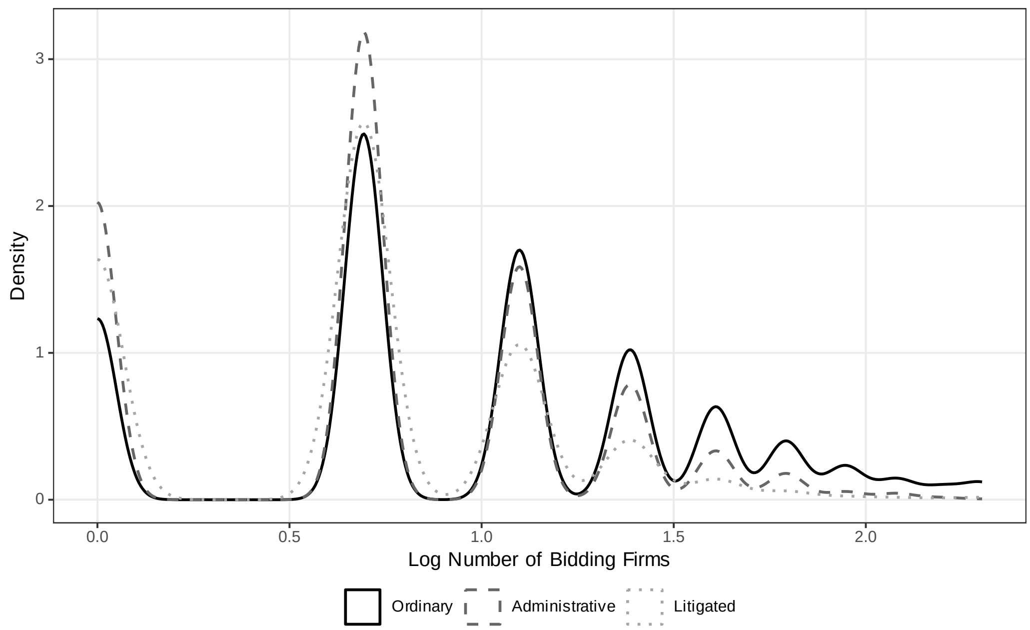


Figure A.6: Distribution of Log Number of Bidding Firms by Purchase Type

procurement failure, the higher the observed success rate, as officials accept less favorable terms to avoid sanctions.

Figure A.9 plots mean log negotiated prices over time for ordinary and urgent purchases separately. The figure reveals two patterns. First, the price gap between urgent and ordinary purchases is persistent throughout the sample period, ruling out the possibility that the regression estimates are driven by a specific subperiod. Second, both series exhibit common trends, supporting the identification assumption that time-varying confounders do not differentially affect the two purchase types.

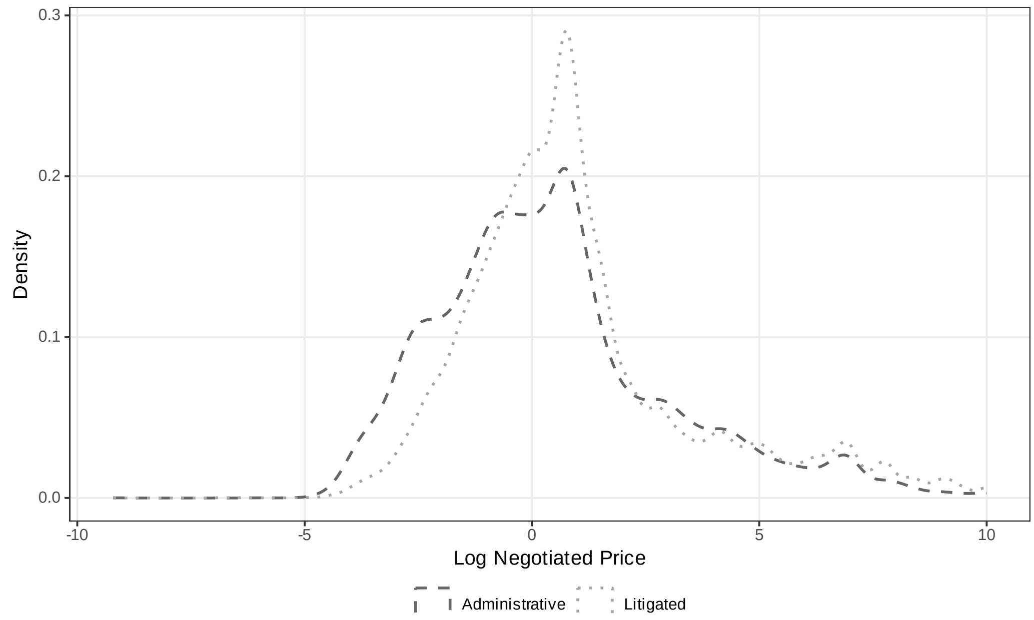


Figure A.7: Distribution of Log Negotiated Price: Administrative vs. Litigated (Urgent Only)

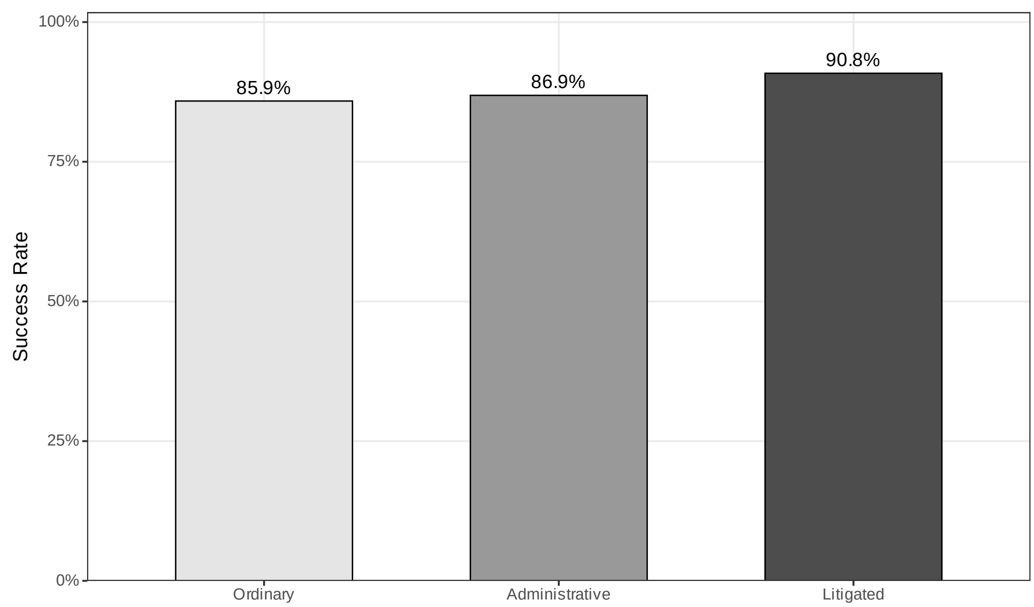


Figure A.8: Tender Success Rate by Purchase Type

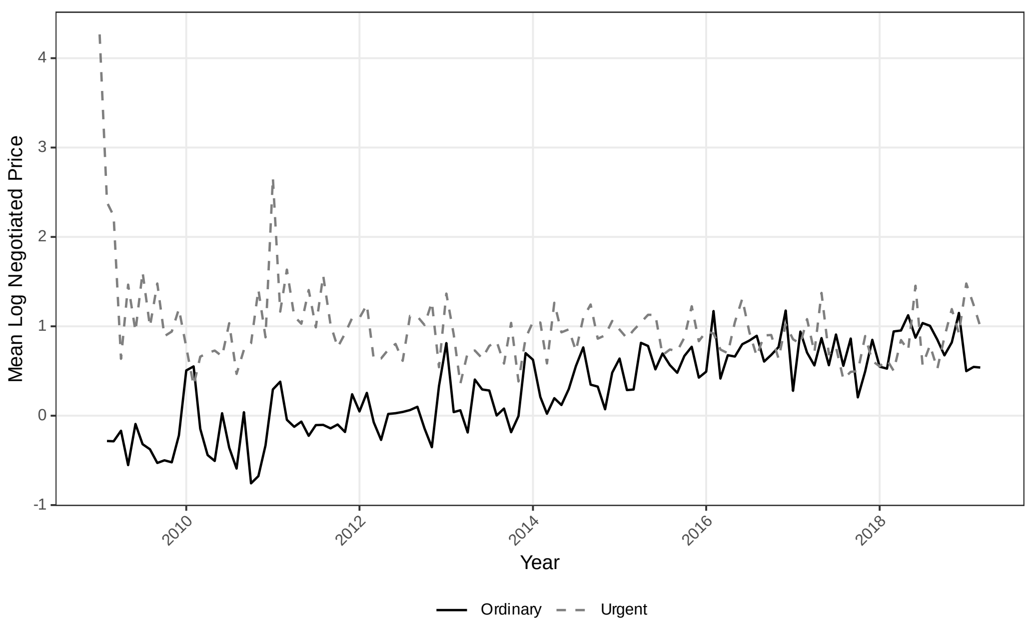


Figure A.9: Mean Log Negotiated Price Over Time: Ordinary vs. Urgent

A.4 Administrative Demand: Descriptive Evidence

This section presents geographic descriptive evidence on the distribution of administrative demand across municipalities in São Paulo, complementing the litigation maps in Section 2. Administrative purchases—urgent procurements initiated through internal government requests rather than court orders—share the same planning constraints as litigated purchases but carry no sanctions for procurement failure. The spatial patterns documented below confirm that administrative and litigated demand originate from similar geographic areas, supporting the institutional comparability that underlies the "under the gun" identification strategy.

Figure A.10 maps administrative purchases per 1,000 inhabitants across municipalities. The geographic distribution closely mirrors the litigation pattern shown in Figure A.1, with higher per capita rates in municipalities that host PBUs and in the interior of the state. This spatial overlap is consistent with both channels responding to the same underlying health needs.

Figure A.11 shows the location of PBUs that process administrative purchases, with marker size proportional to the volume of transactions. The distribution mirrors the PBU map for litigated purchases (Figure A.2), confirming that the same institutional infrastructure handles both procurement channels.

Figure A.12 provides a visual comparison of administrative and litigated purchases, highlighting that the procurement process is identical across both channels—the only distinguishing feature is the threat of judicial sanctions

Figure A.10: Administrative Purchases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors' elaboration based on BEC procurement data and IBGE population estimates.

in litigated cases. This institutional parallel is the foundation for the "under the gun" identification.

Figure A.13 maps the total number of administrative purchases by municipality. As with litigation, administrative demand is concentrated in a small number of urban centers, particularly the São Paulo metropolitan area and regional hubs in the state's interior.

Figure A.14 shows the share of administrative purchases as a proportion of total purchases (administrative plus litigated) by municipality. The substantial variation across municipalities suggests that the relative importance of administrative versus litigated demand reflects local institutional factors—

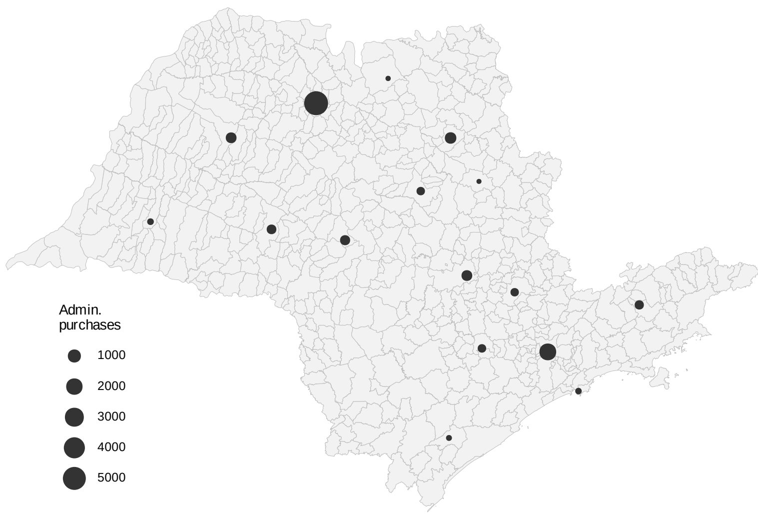


Figure A.11: PBUs Handling Administrative Purchases (Size ∝ Volume)
Source: Authors' elaboration based on SES/SP administrative records.

such as the presence of administrative request mechanisms and the propensity of local courts to grant injunctions—rather than a uniform statewide pattern.

	Administrative	Litigated
Origin	SES/SP request	Court order
Source of funds	Diverted budget	Diverted budget
Quantity	Small	Small
Delivery time	Short	Short
Threat of punishment	None	Potential punishment

Figure A.12: Comparison of Administrative and Litigated Purchases
Source: Authors' elaboration.

Figure A.13: Total Administrative Purchases by Municipality in São Paulo (2009–2019)
Source: Authors' elaboration based on BEC procurement data.

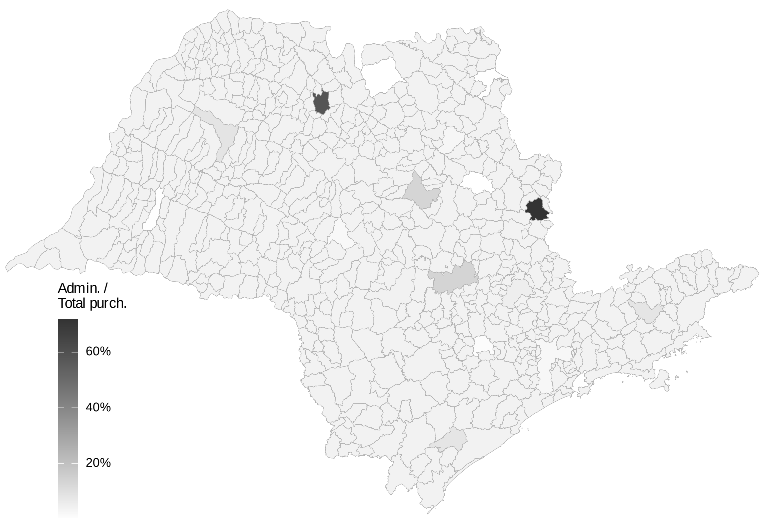
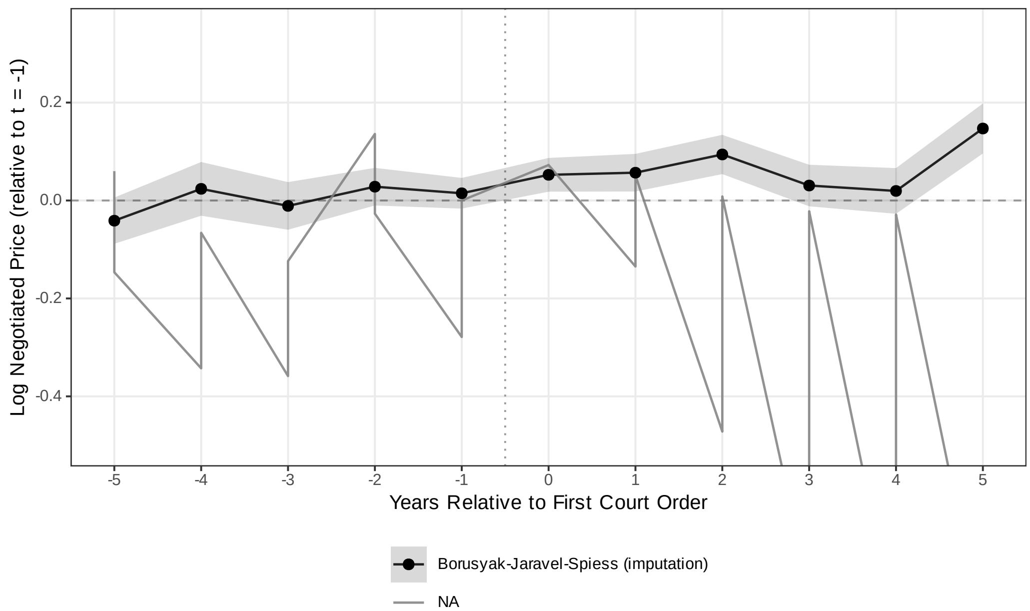


Figure A.14: Administrative Purchases as Share of Total Urgent Purchases by Municipality
Source: Authors' elaboration based on BEC procurement data.

A.5 Event Study: First Court Order (Honest DiD)

To address concerns about pre-trends and heterogeneous-treatment-effect contamination in the TWFE event-study design, we re-estimate the dynamic response of log negotiated prices to the first court order for each item using two contemporary estimators: the imputation estimator of Borusyak et al. (2024) (BJS) and the ATT(g, t) estimator of Callaway and Sant'Anna (2021) (CS). The panel is at the item-year level with winner-only observations; treatment is defined as the calendar year of the first litigated purchase for each item; the BJS specification uses item + year fixed effects fit on untreated observations to impute the counterfactual. The CS specification uses never-litigated items as the control group.

BJS is our preferred honest specification because it uses pre-treatment observations of eventually-treated items to fit the counterfactual model, avoiding the selection problem that haunts the CS nevertreated comparison. The pattern is consistent with the cross-sectional estimates in the main text: after a court order, the same item is systematically more expensive than it was in its own pre-period, with no detectable pre-trend that could confound the interpretation.



referred (shaded 95% CI). CS with never-treated control is noisier because never-litigated items are a systematically different comparison group.

Figure A.15: Log Negotiated Price Around First Court Order: Honest DiD Estimators

Notes: Item-year panel, winner-only observations. BJS (Borusyak et al. 2024) shown with shaded 95% confidence interval; CS (Callaway and Sant'Anna 2021) with never-treated control group; TWFE shown only as a pre-reform benchmark. The BJS pre-period coefficients are economically small (all < 0.041 in absolute value) and statistically indistinguishable from zero, consistent with parallel trends; post-treatment coefficients rise monotonically from $+0.052$ at $t = 0$ to $+0.147$ at $t = +5$. The CS estimator with never-treated control is substantially noisier because items that never attract a court order are a systematically different comparison group (they tend to be routine, well-stocked items); HonestDiD (Borusyak et al. 2024-compatible sensitivity in the companion CSV) confirms that the CS point estimate at $t = 0$ is not robust to parallel-trend violations of moderate magnitude.

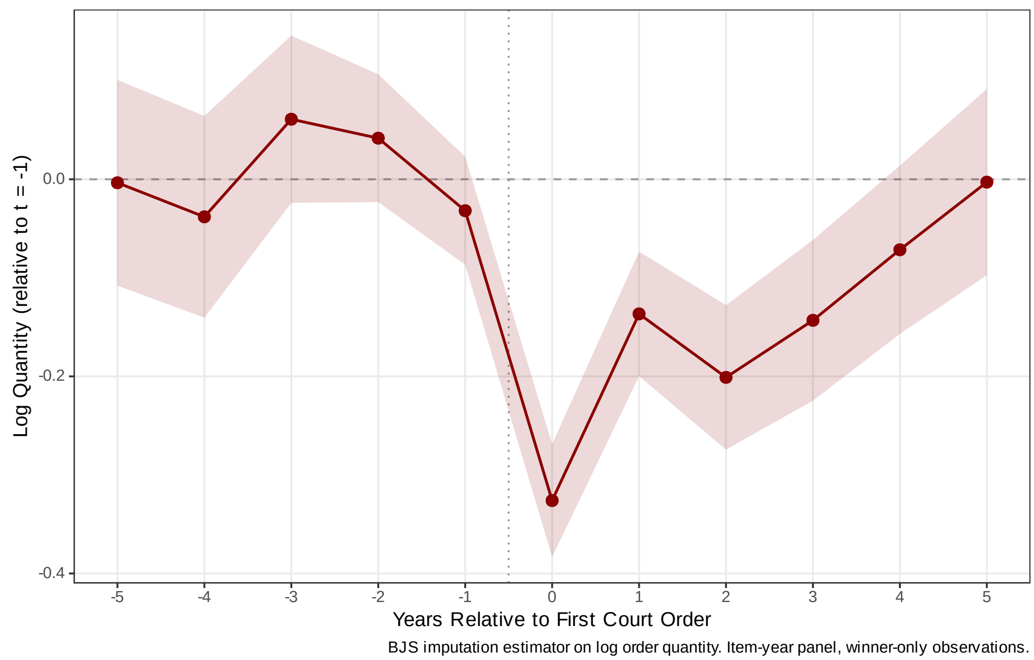


Figure A.16: Log Order Quantity Around First Court Order: Borusyak-Jaravel-Spiess Imputation

Notes: Item-year panel, winner-only observations. BJS imputation estimator on log order quantity. Within the same item, order quantity drops sharply at the year of the first court order (-27.8% at $t = 0$) and recovers thereafter (-0.3% at $t = +5$). The dynamic pattern documents the C1 fragmentation channel: judicial pressure forces smaller orders in real time, on the same item.

A.6 Heterogeneity

We examine heterogeneity in the urgency premium along four dimensions. Tables A.14–A.17 report results from split-sample regressions and interaction models using our preferred specification (item + year + PBU FE).

Table A.14: Heterogeneity: SUS Component

	Split Sample		Interaction
	Specialized	Basic SUS	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.005	0.030*	-0.073
	(0.046)	(0.016)	(0.097)
Urgent × Basic SUS			0.159
			(0.115)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	45,048	151,835	196,883
Within R^2	0.000	0.000	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

SUS component (Table A.14).. The urgency premium on negotiated prices is concentrated among items in the basic SUS component (common medications): the coefficient is 0.030 ($p < 0.10$) for basic items and an insignificant 0.005 for specialized items, suggesting that procurement disruption is more consequential for items where established supply chains normally deliver better prices.

Time period (Table A.15).. The urgency premium is larger in the later period (2014–2019) than in the earlier period (2009–2013), consistent with the

Table A.15: Heterogeneity: Time Period

	Split Sample		Interaction
	Early	Late (2014+)	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.074***	0.045**	0.259***
	(0.025)	(0.018)	(0.035)
Urgent × Late Period			-0.339***
			(0.043)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	82,658	113,936	196,883
Within R^2	0.000	0.000	0.005

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

growing volume and complexity of health litigation over time.

Market competition (Table A.16).. The urgency premium is three times larger in competitive markets (0.143 vs. 0.046), with a significant interaction (0.226, $p < 0.01$). In concentrated markets, fewer bidders leave less room for urgency to erode bargaining power; in competitive markets, the loss of planning time has the largest marginal impact.

PBU size (Table A.17).. The urgency premium is slightly larger for purchases made by smaller PBUs (below-median transaction volume), consistent with smaller units having less experience managing urgent procurement.

Table A.16: Heterogeneity: Market Competition

	Split Sample		Interaction
	Low Comp.	High Comp.	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.046***	0.143***	0.026
	(0.016)	(0.034)	(0.018)
Urgent × High Competition			0.226***
			(0.044)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	125,950	61,313	187,264
Within R^2	0.000	0.001	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

Table A.17: Heterogeneity: PBU Size

	Split Sample		Interaction
	Small PBU	Large PBU	Full Sample
	(1)	(2)	(3)
Urgent Purchase	0.023	0.062***	-0.000
	(0.026)	(0.018)	(0.049)
Urgent × Large PBU			0.070
			(0.054)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	31,902	164,511	196,883
Within R^2	0.000	0.000	0.000

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** p<0.01, ** p<0.05, * p<0.1.

A.7 Regex Classifier Validation

The purchase-type indicator is constructed by a regular-expression algorithm applied to public tender notices, which by Brazilian procurement law must reference any underlying court order. To document classifier accuracy, we draw a stratified random sample of 500 tender-notice subjects (approximately 167 per predicted class) and hand-label each as ordinary, administrative, or litigated. Hand-labeling of the stratified sample is in progress; F1 diagnostics will be reported in the replication archive accompanying this paper. The regex's literal-reference design implies misclassification, if any, is near-symmetric across the three classes: the algorithm relies on textual markers that are present or absent independently of the price outcomes we estimate, so misclassification attenuates rather than biases our estimates.[4]

A.8 Selection Bounds, Reconciliation, and Diagnostic Robustness

This section collects the selection-bound, reconciliation, and diagnostic robustness exhibits that support the main-text claims in Sections 4 and 5.

Manski-Lee bounds on the UTG.. Table A.18 reports the bounded UTG coefficient under monotone selection. Trimming admin observations within item×year×PBU strata where the admin sample exceeds the litigated count yields a bounded gap in $[15.9\%, 21.1\%]$, against a naive coefficient of 29.5%.

[4] The stratified sample is reproduced from `output/validation/validation_sample.csv`, generated by `analysis/33_regex_validation_sample.R`; F1 statistics are emitted by `34_regex_validation_f1.R` when the labeled CSV is finalized.

Table A.18: Manski-Lee bounds on the under-the-gun coefficient.

	log negotiated price		implied
Specification	Coef. on Admin	SE	Lit-vs-Admin (%)
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530
Lee lower bound (admin top-trim)	-0.192	0.095	15.924
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123

Trimming applied within item*times*year*times*PBU strata where the admin sample exceeds the li

Average trimming intensity is 26.9% of admin observations across 12,038 of 44,654 strata.

Heckman selection model (sensitivity).. Table A.19 reports a parametric Heckman correction with the same first-stage probit (pre-period log reference price; pre-period mean quantity; pre-period mean number of bidders; pre-period order count—the exclusion restriction; SUS-formulary indicator). The first-stage probit on $N = 5{,}351$ observations confirms that pre-period reference price and SUS status both enter negatively and significantly. The second-stage Admin coefficient is 0.817 (SE 4.278); the inverse Mills ratio is collinear with the item, year, and PBU fixed effects, so the second-stage point estimate is uninformative. We treat this as a sensitivity diagnostic that flags identification fragility, not as a competing point estimate, and rely on the Lee bound as the primary selection-corrected object.

Both-types-cell representativeness.. Table A.20 compares the 5,581 item×year-month cells that contain both administrative and litigated urgent purchases (the within-cell variation that identifies the tightest specification) with the

Table A.19: Heckman selection-corrected UTG estimate.

Term	Estimate
First stage: pre-period log ref price	-0.051
First stage: pre-period mean qty	0.000
First stage: pre-period mean firms	0.065
First stage: pre-period n orders (excl)	0.000
First stage: SUS formulary	-0.265
Second stage: Admin coef	0.817
Second stage: inverse Mills ratio	-0.653
Second stage: implied lit-over-admin (%)	-55.841
Heckit ML: ρ	1.000

First stage: probit of admin on pre-period item characteristics. pre_n_orders is excluded from t

11,972 singleton cells. Both-types cells systematically have lower mean log reference and negotiated prices, larger quantities, slightly fewer bidders, and a higher SUS-formulary share. Bounded UTG estimates therefore speak to the more liquid, formulary-anchored items in the urgent panel; we read this as a generalizability boundary, not as a threat to within-cell identification.

Wild cluster bootstrap.. Table **??** reports Rademacher wild cluster bootstrap inference on the UTG coefficient (999 replicates, PBU clustering). The preferred-FE Admin coefficient is significant under both asymptotic and bootstrap inference ($p_{\text{boot}} = 0.0080$, 95% CI in log-points $[-0.469, -0.039]$). The item×year-month FE specification is borderline ($p_{\text{boot}} = 0.0390$), as expected given the thin within-cell sample.[5]

[5] Implementation: manual Rademacher implementation (the `fwildclusterboot` package is unavailable for R 4.5); see `analysis/44_wild_bootstrap.R`. Restricted-null residuals weighted by cluster-level Rademacher draws; bootstrap t-distribution constructed

Table A.20: Representativeness of both-types cells: cells with both administrative and litigated urgent purchases (5,581 cells, 31.8%) vs. singleton cells (11,972).

Variable	Both-types cells	Singleton cells
Log reference price	1.233	1.761
Log negotiated price	0.703	1.326
Log quantity	6.560	6.324
N. bidding firms	2.181	2.404
SUS formulary share	0.867	0.725
Reverse-auction (pregao) share	0.955	0.952

Means computed at the observation level within each cell type. p-values from two-sample t-tests.

Rambachan-Roth honest sensitivity on BJS event-study.. Figure A.17 reports the Borusyak-Jaravel-Spiess imputation event-study coefficients with a Rambachan-Roth honest sensitivity overlay. Pre-period maximum absolute coefficient is 0.041; the $t = 0$ effect is 0.052 (SE 0.018); the breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero is $M = 0.018$. The $t = 0$ effect is significant under parallel trends but does not survive the linear extrapolation of the observed pre-period maximum—we honor this in the main text by treating the cross-sectional fixed-effects estimate as primary and the BJS event study as supportive, not as a stand-alone identification claim.

Reconciliation table.. Table A.21 decomposes the observed log-price gap (admin minus litigated) into the mechanical bulk-discount channel (−32.8%, admin cheaper from order-size differences alone), the within firm-buyer-item

from 999 replicates.

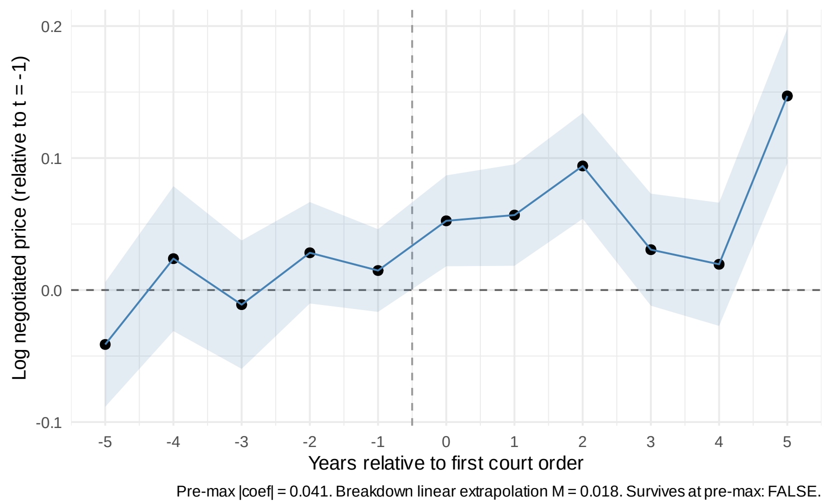


Figure A.17: BJS event study around first court order with Rambachan-Roth sensitivity diagnostic.

offset (3.5%, NULL), and the supplier composition residual (10.9%). Same firms price the same items the same way regardless of regime. The composition residual—positive in the admin-minus-lit direction—reflects that the litigated supplier mix is, conditional on quantity, less expensive per unit than the administrative supplier mix; the bulk-discount disadvantage of small lit orders dominates, producing the net lit-over-admin observed gap.

Table A.21: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log-points (admin minus lit)
Observed UTG (admin minus lit, log price)	-0.259
Mechanical C1: bulk-discount × qty gap	-0.397
Within firm-buyer-item offset	0.035
Supplier composition residual	0.104

Identity: observed UTG = mechanical C1 + within-firm offset + composition residual. Within-fi

Table A.22: Selection-bias-corrected welfare bound on the procurement cost of judicial enforcement.

Component	Value
Bounded UTG (Lee midpoint)	18.5%
Lee lower bound	15.9%
Lee upper bound	21.1%
Admissibility share (calibration)	50%
Annual litigated spend, SP	\$300 M / yr
Welfare bound (Lee midpoint x admissibility x spend)	\$27.8 M / yr
Welfare bound (Lee lower x admissibility x spend)	\$23.9 M / yr
Welfare bound (Lee upper x admissibility x spend)	\$31.7 M / yr

Bounded UTG: Manski-Lee bounds from Table A.18. Admissibility share: calibration anchor for

Welfare bound.. Table A.22 reports the selection-bias-corrected welfare bound. Applying the Lee-midpoint UTG of 18.5% to \$300 M of annual litigated spending in São Paulo, with an admissibility-share calibration of 50.0%, yields \$27.8 M per year on the per-unit-price margin. The Lee-endpoint range is $[\$23.9\ M, \$31.7\ M]$. This is a per-unit-price bound and excludes welfare losses from constrained order sizes, reduced bidder participation, and officials' search time diverted from ordinary procurement; the headline number understates the total cost.